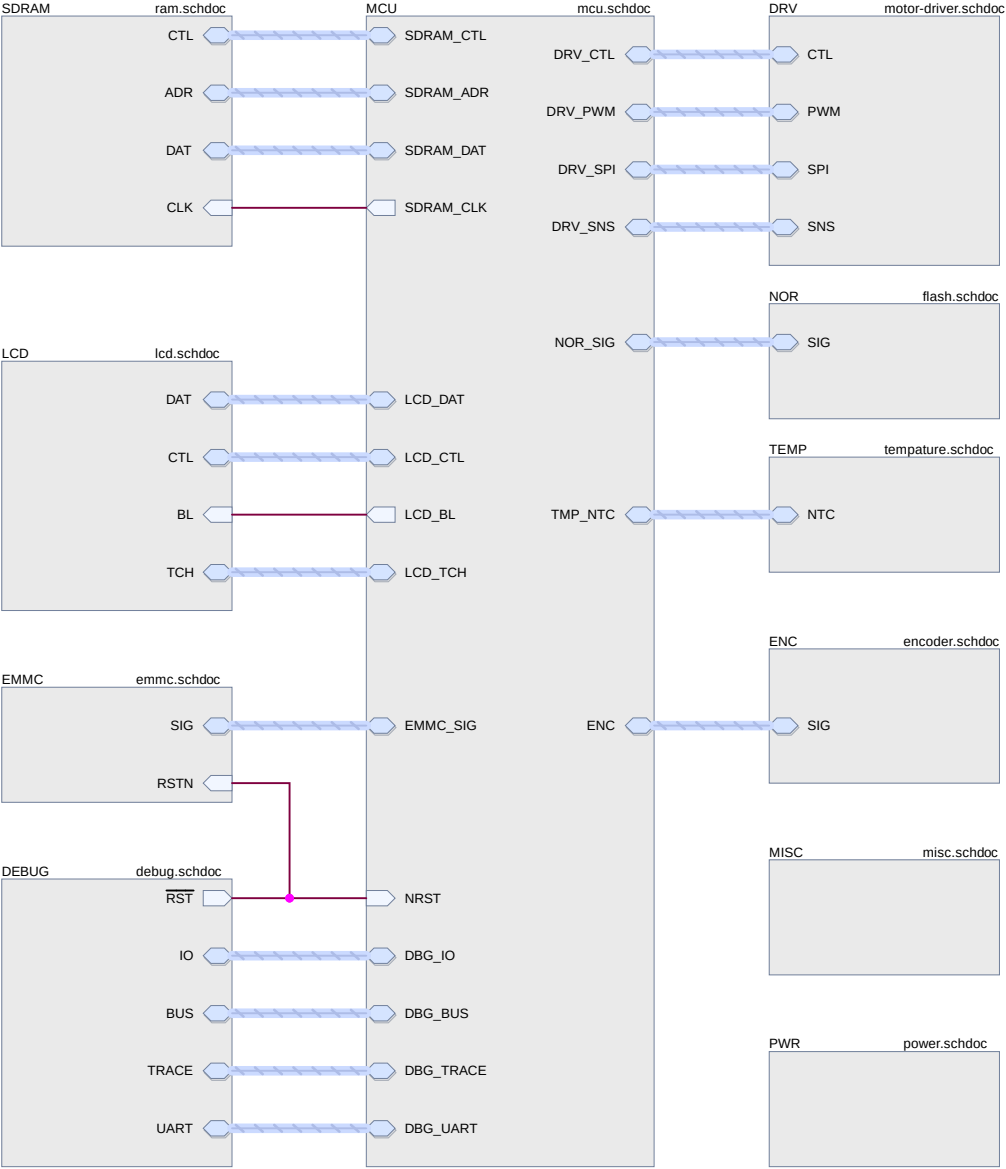
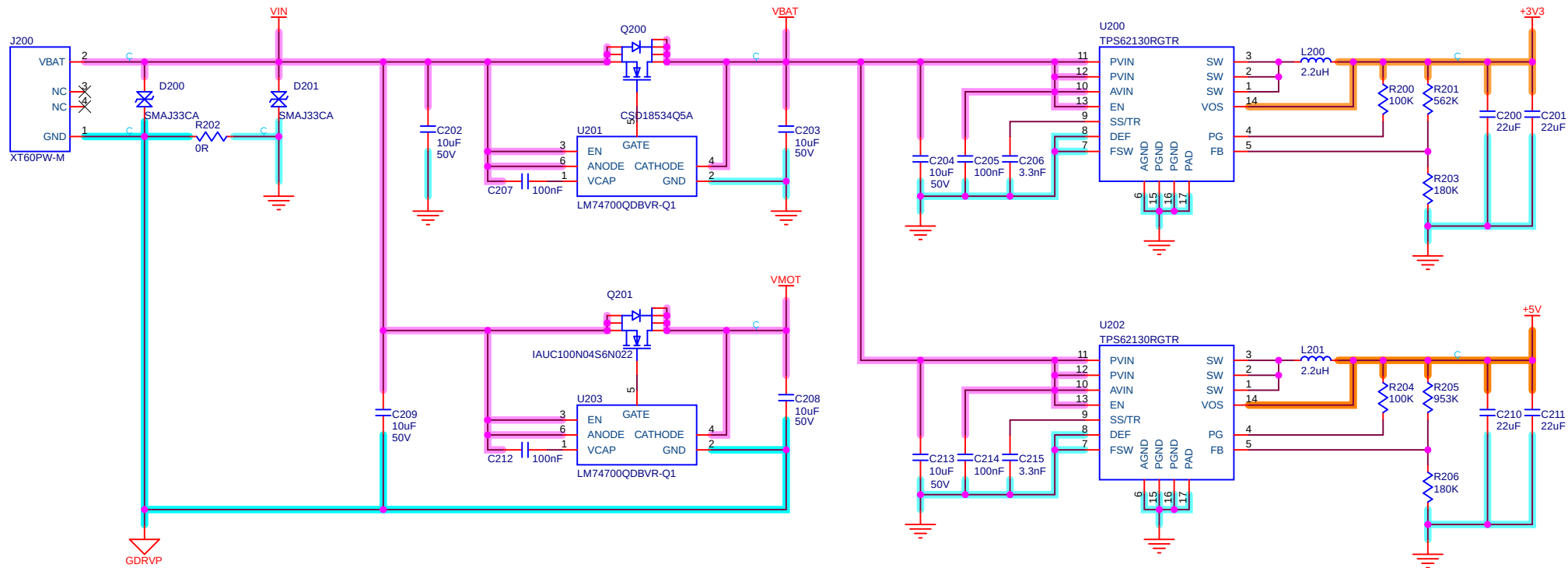
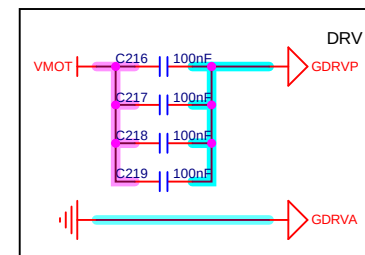
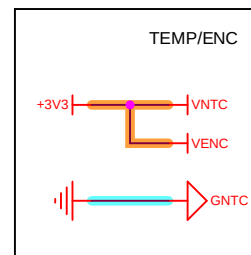
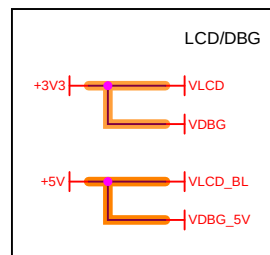
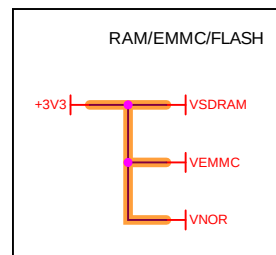
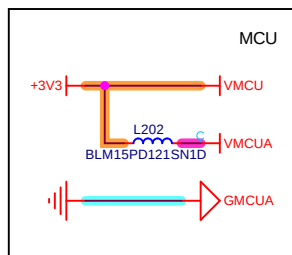






Here is only the network name conversion, no single-point grounding/supplying is required. Debug interface is 3.3V, and 5V is reserved for future use. All components shall close to the module or IC.



PWR
Power filter and rails

v1.0.0 GPLv3

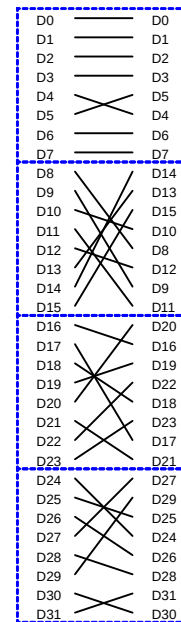
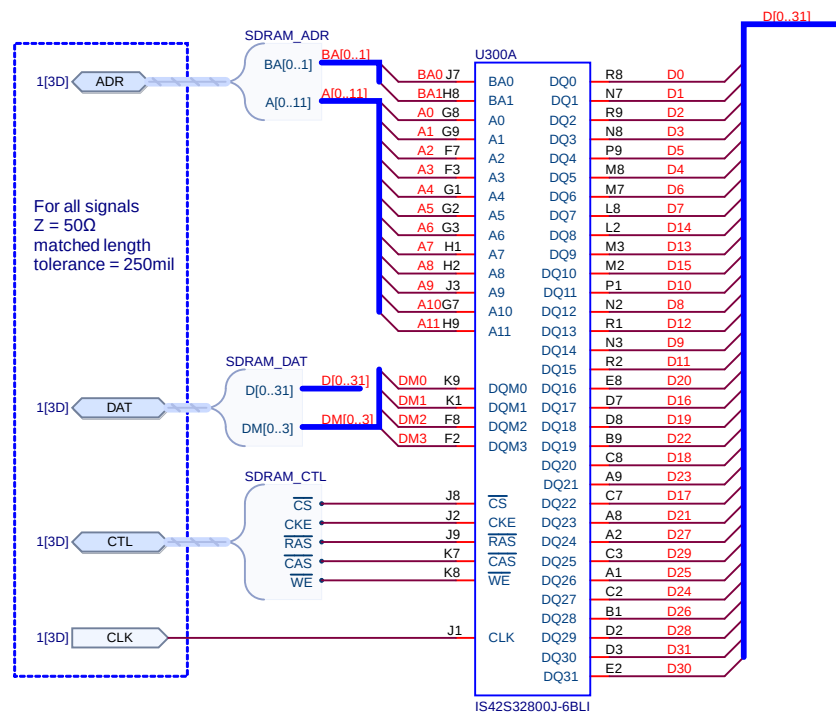
DRAWN 10.25.2025
DRAWNBY miaow
CHECKED BCY
SHEETS 14
SHEETS_NO 2

MIAOW FOC BOARD

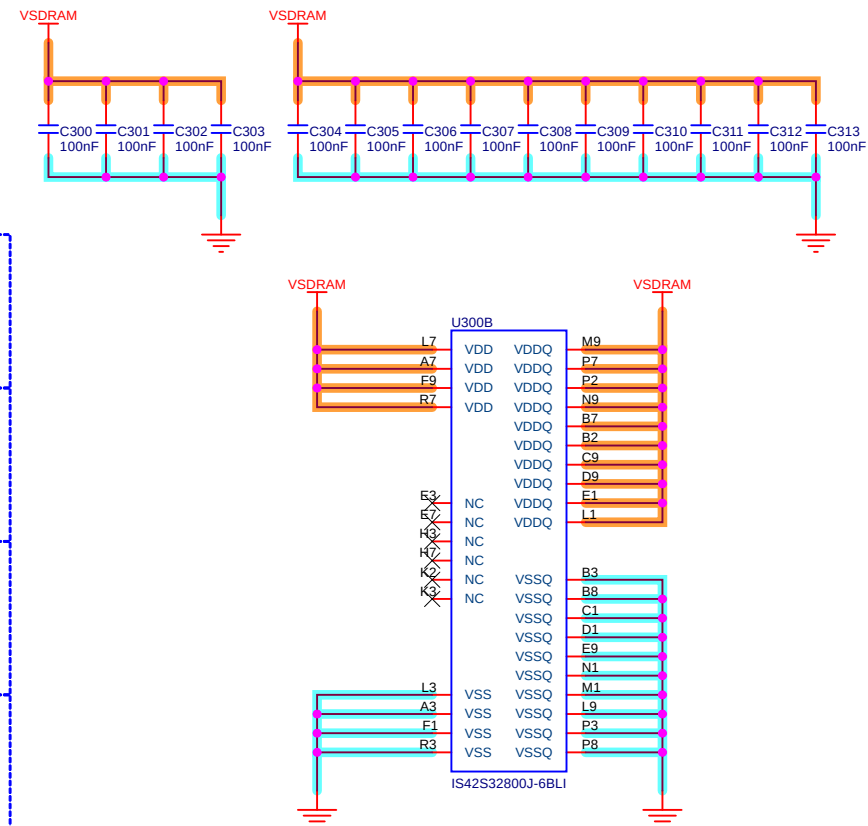
MAIN_PCB AG

GPLv3

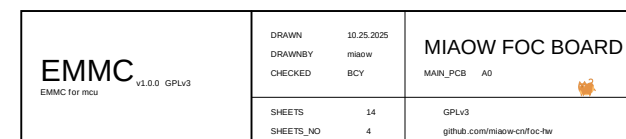
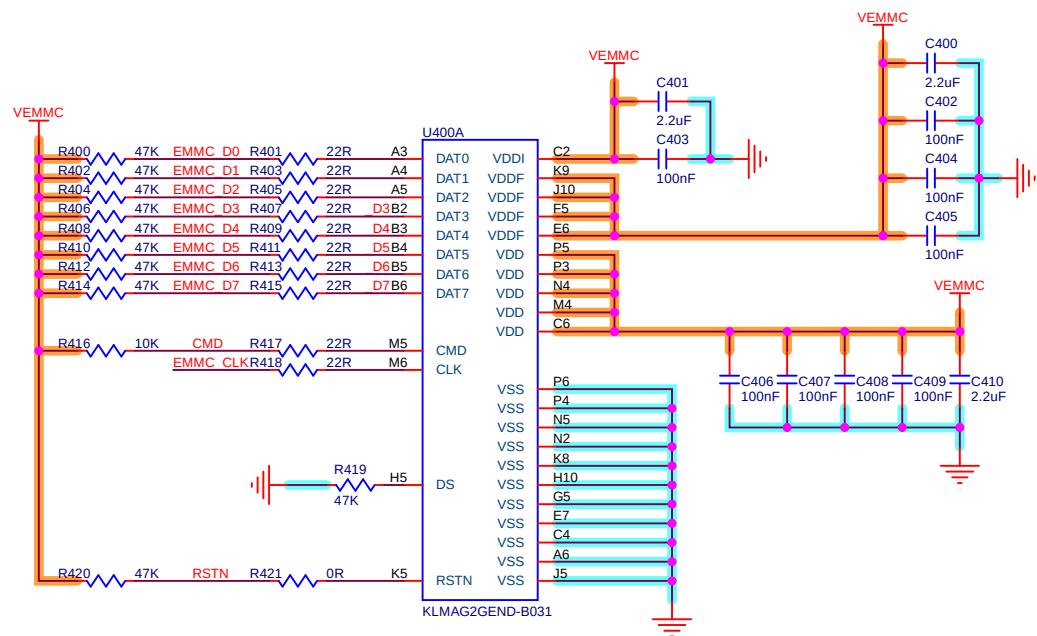
github.com/miaow-cn/focthw

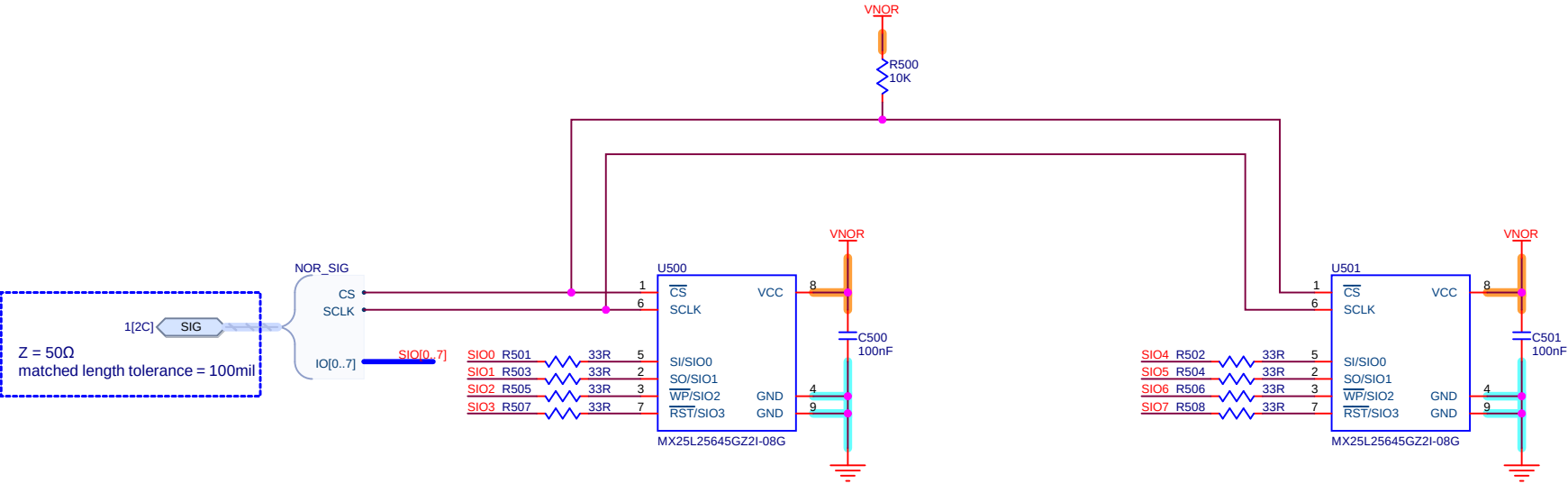


Data lines in the same
byte group are
remapped for routing

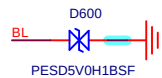
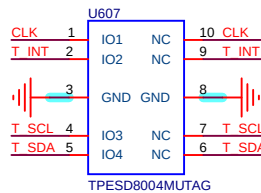
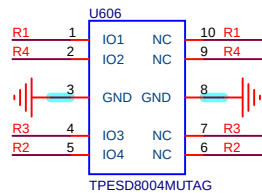
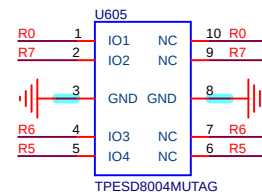
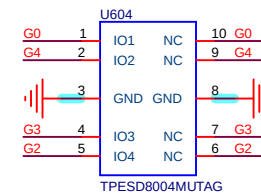
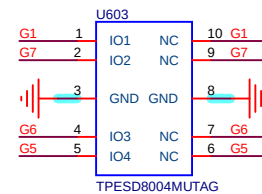
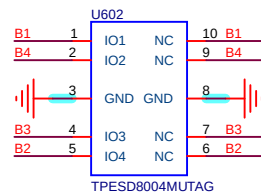
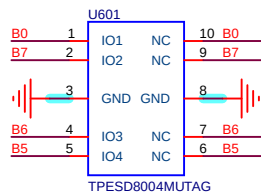
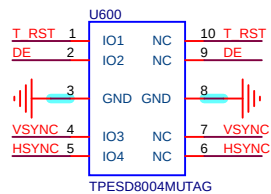
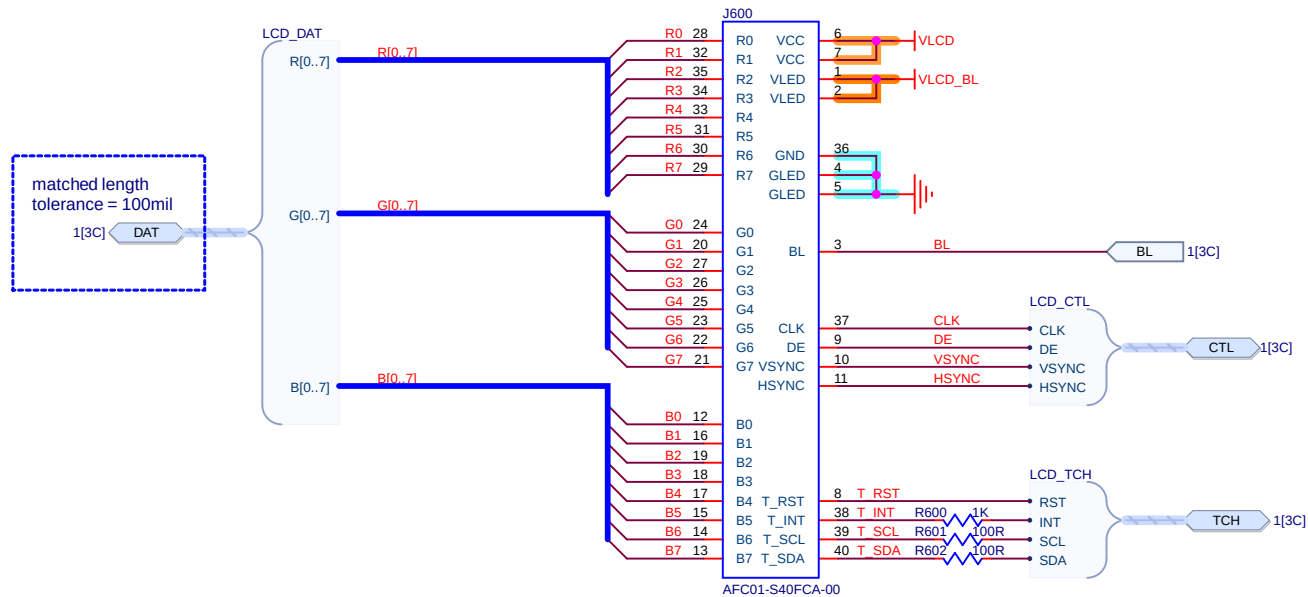


RAM SDRAM for mcu	v1.0.0 GPLv3	DRAWN	10.25.2025	MIAOW FOC BOARD	
		DRAWNBY	miaow	MAIN_PCB	A0
		CHECKED	BCY		
		SHEETS	14	GPLv3	
		SHEETS_NO	3	github.com/miaow-cn/foctw	

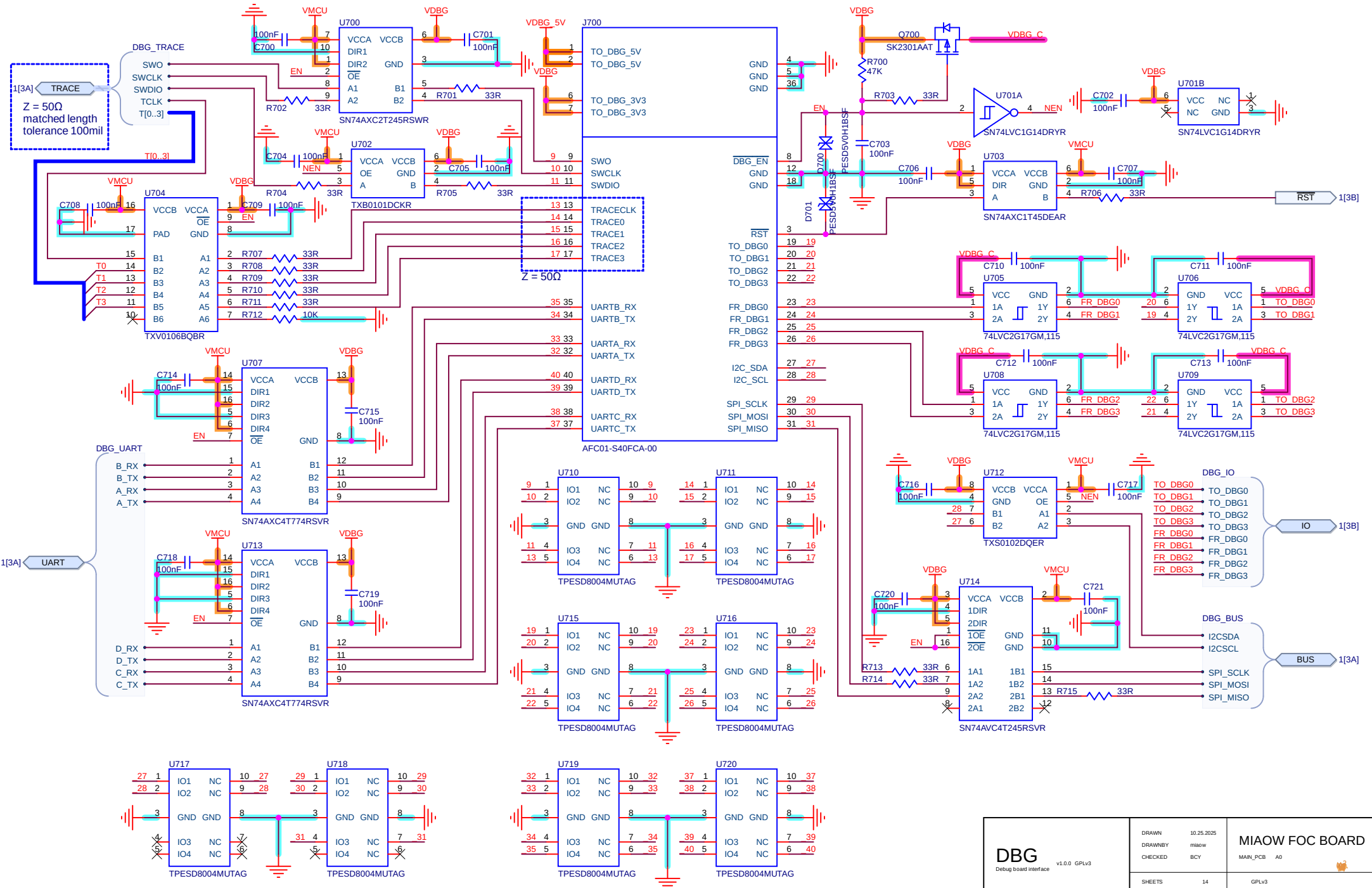




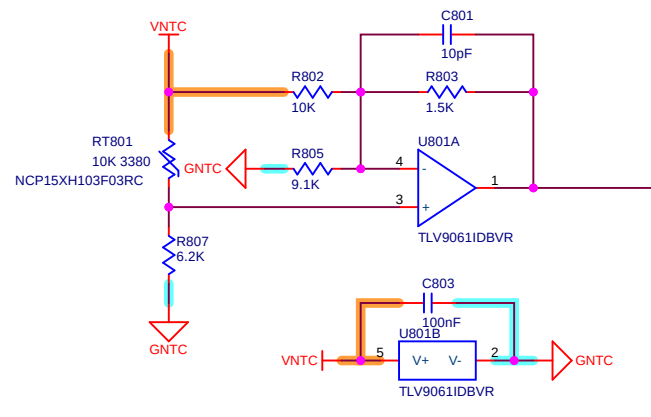
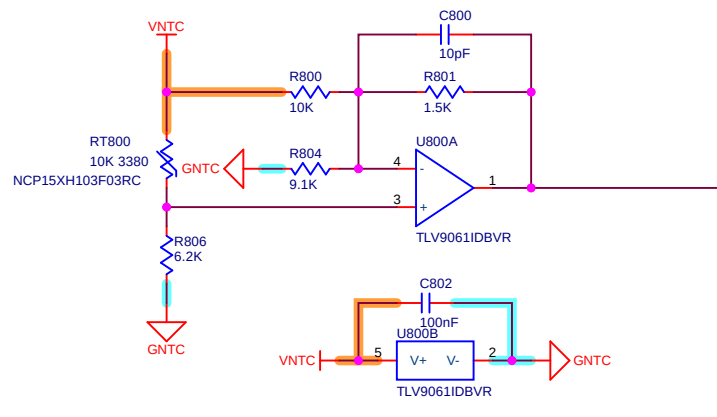
FLASH NOR flash for mcu	DRAWN	10.25.2025	MIAOW FOC BOARD
	DRAWNBY	miaow	
	CHECKED	BCY	
SHEETS		14	GPLV3
SHEETS_NO		5	github.com/miaow-cn/oc-hw



LCD RGB LCD interface v1.0.0 GPLv3	DRAWN10.25.2025		MIAOW FOC BOARD	
	DRAWNBYmiaow		MAIN_PCB A0	
	CHECKEDBCY			
	SHEETS14			
SHEETS_NO6				
			GPLv3	
			github.com/miaow-cn/fof-hw	

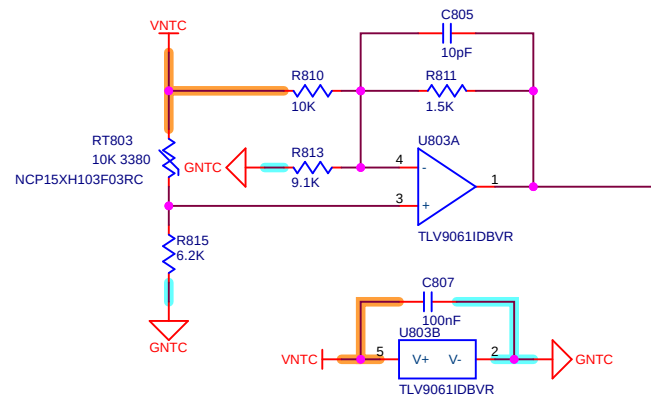
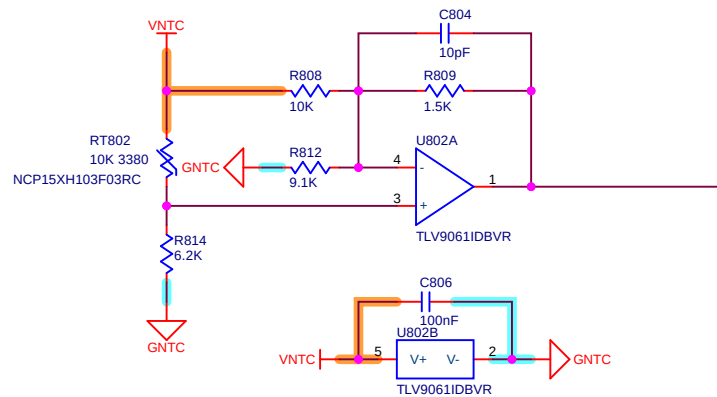
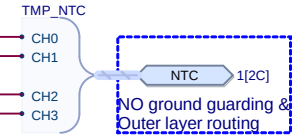


DBG Debug board interface v1.0.0 GPLv3	DRAWN	10.25.2025	MIAOW FOC BOARD MAIN_PCB A0
	DRAWNBY	miaoow	
	CHECKED	BCY	
	SHEETS	14	
	SHEETS_NO	7	GPLv3 github.com/miaoow-cn/fooc-hw

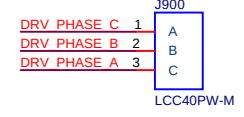
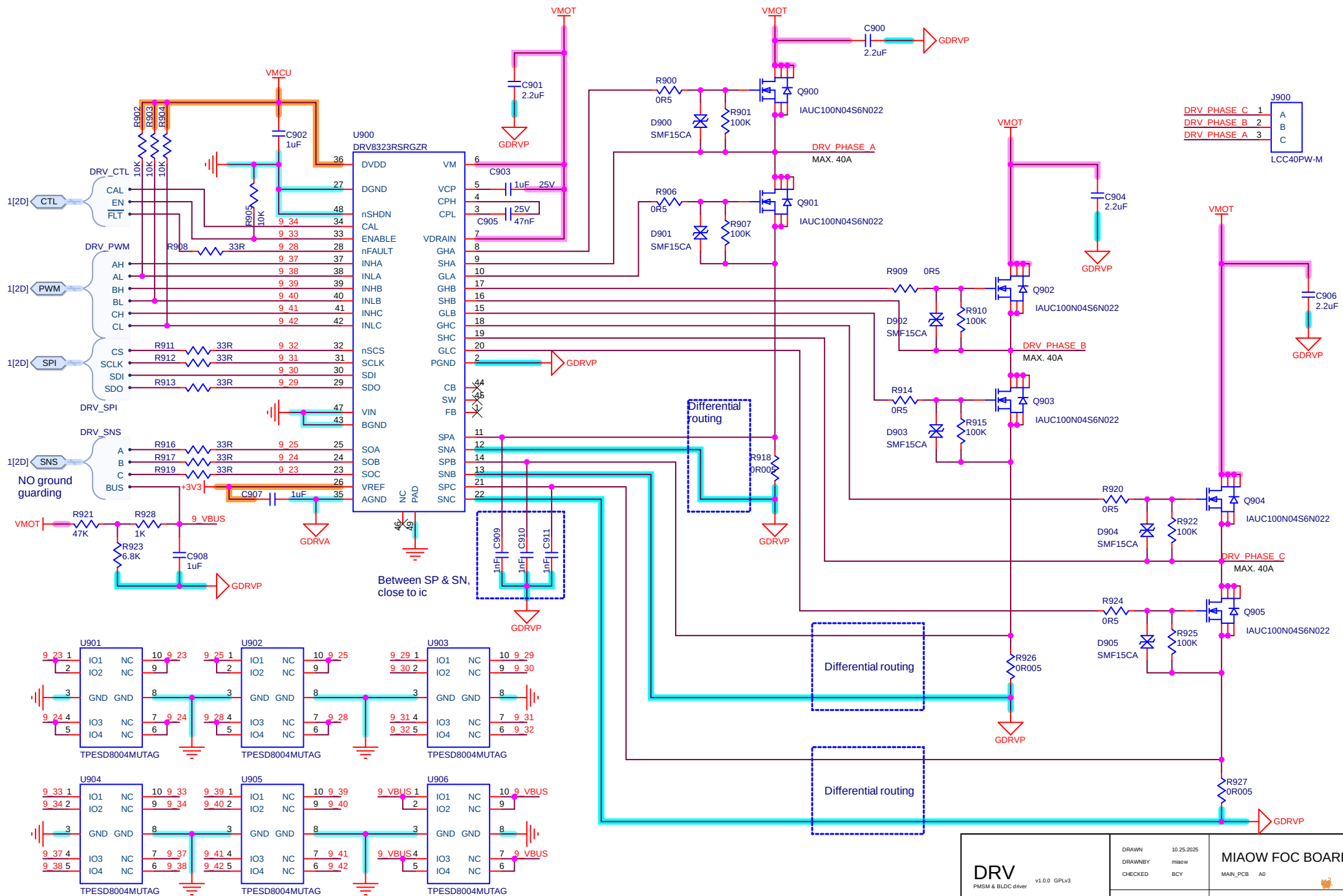


gain and offset is calculated with 3.3V power supply

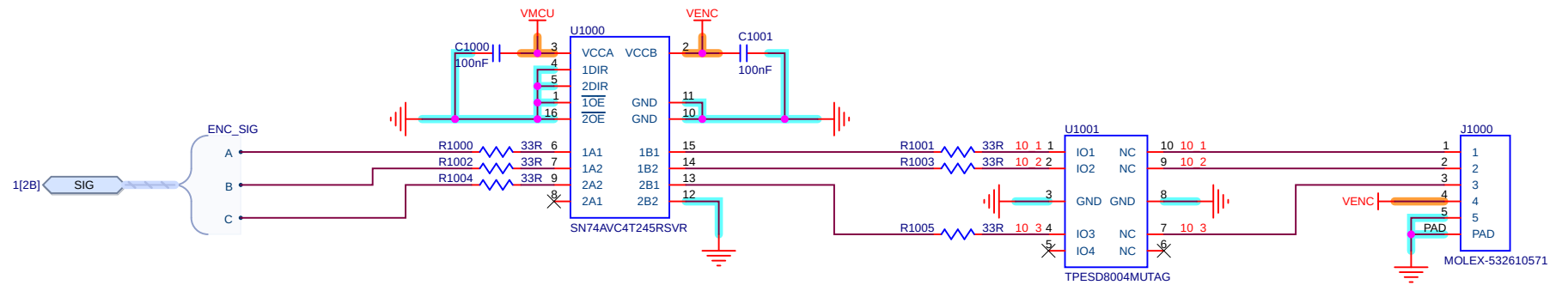
-20 - 100 °C
42.506 KΩ - 0.974 KΩ
OUT_min = 0.057325V
OUT_max = 3.254864V



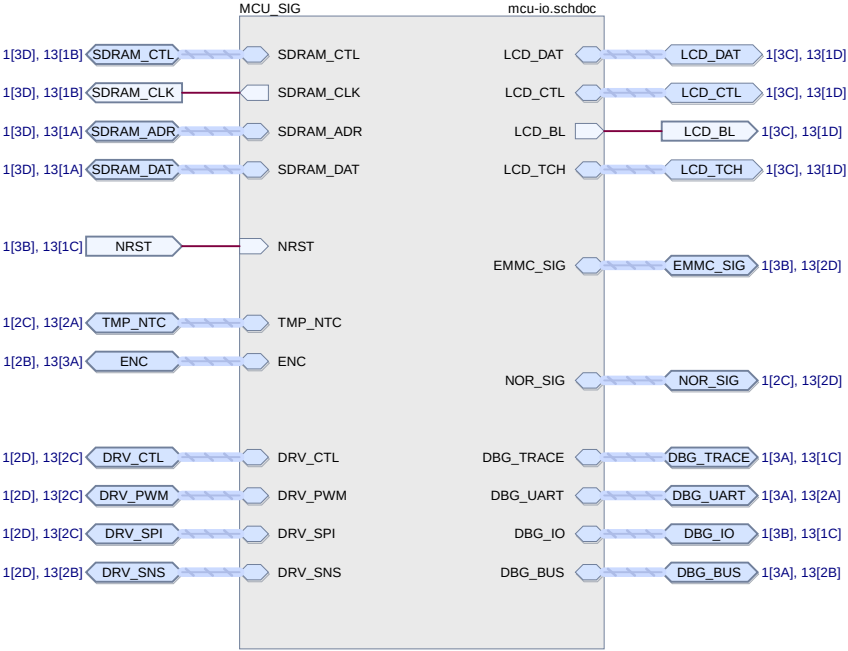
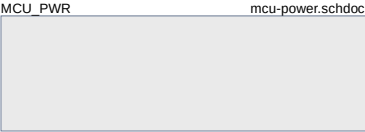
TEMP NTC temperature sensor v1.0.0 GPLv3	DRAWN	10.25.2025	MIAOW FOC BOARD MAIN_PCB AG
	DRAWNBY	miaoow	
	CHECKED	BCY	
	SHEETS	14	GPLv3
	SHEETS_NO	8	github.com/miaoow-cn-foc-hw



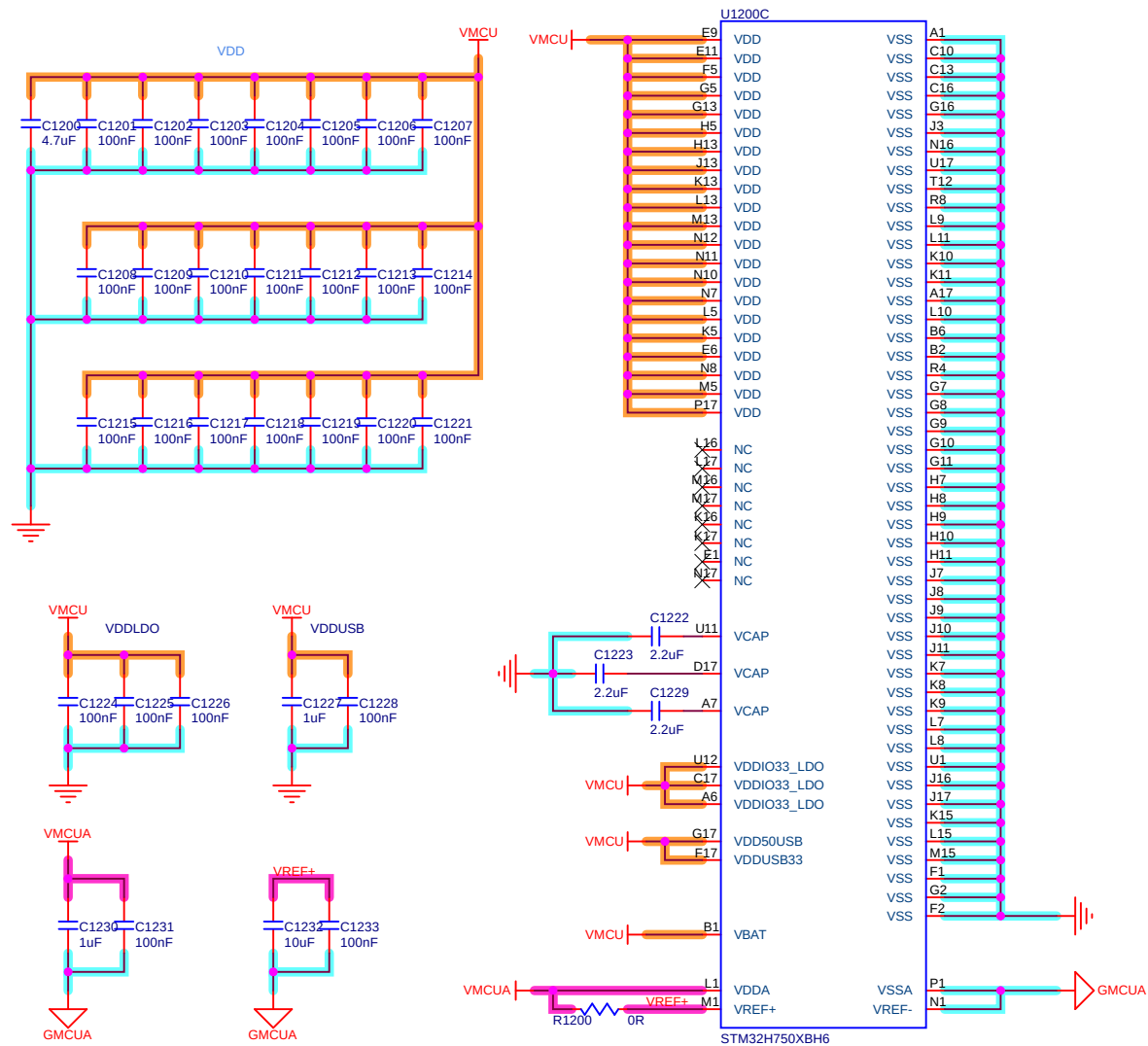
DRV PMSM & BLDC driver v1.0.0 GPLv3	DRAWN	10.25.2025	MIAOW FOC BOARD MAIN_PCB AG
	DRAWNBY	miaoow	
	CHECKED	BCY	
	SHEETS	14	
	SHEETS_NO	9	GPLv3 github.com/miaoow-cnfoctw



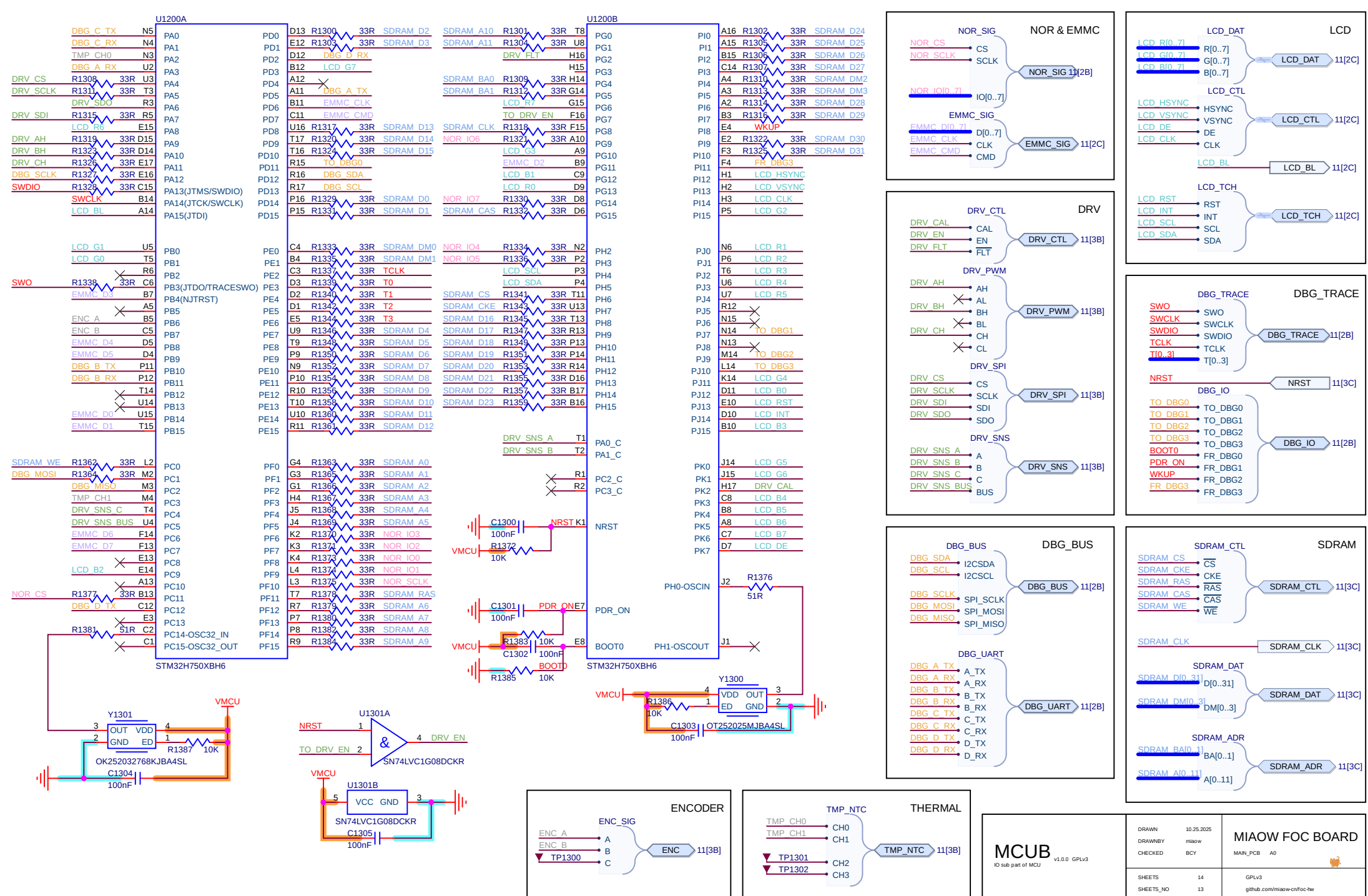
ENC Encoder interface	v1.0.0 GPLv3		DRAWN 10.28.2025	MIAOW FOC BOARD
			DRAWNBY miaow	
			CHECKED BCY	
			MAIN_PCB A0	
		SHEETS 14	GPLv3	
		SHEETS_NO 10	github.com/miaow-cn/pcb-ib	



MCU MCB block diagram	v1.0.0 GPLv3	DRAWN	10.25.2025	MIAOW FOC BOARD
		DRAWNBY	miaow	
		CHECKED	BCY	
		SHEETS	14	GPLv3
		SHEETS_NO	11	github.com/miaow-cn/foctw



MCUA Power subpart of MCU	DRAWN	10.25.2025	MIAOW FOC BOARD MAIN_PCB A0
	DRAWNBY	miaow	
	CHECKED	BCY	
	SHEETS	14	
	SHEETS_NO	12	GPLv3 github.com/miaow-cn/foctw





POSHOLE1400



POSHOLE1401



POSHOLE1402



POSHOLE1403



POSHOLE1404



POSHOLE1405



MARKPOINT1400



MARKPOINT1401



MARKPOINT1402



MARKPOINT1403



BAR1400



BAR1402



BAR1404



BAR1406



BAR1408



VMOT



BAR1401



BAR1403



BAR1405



BAR1407



BAR1409

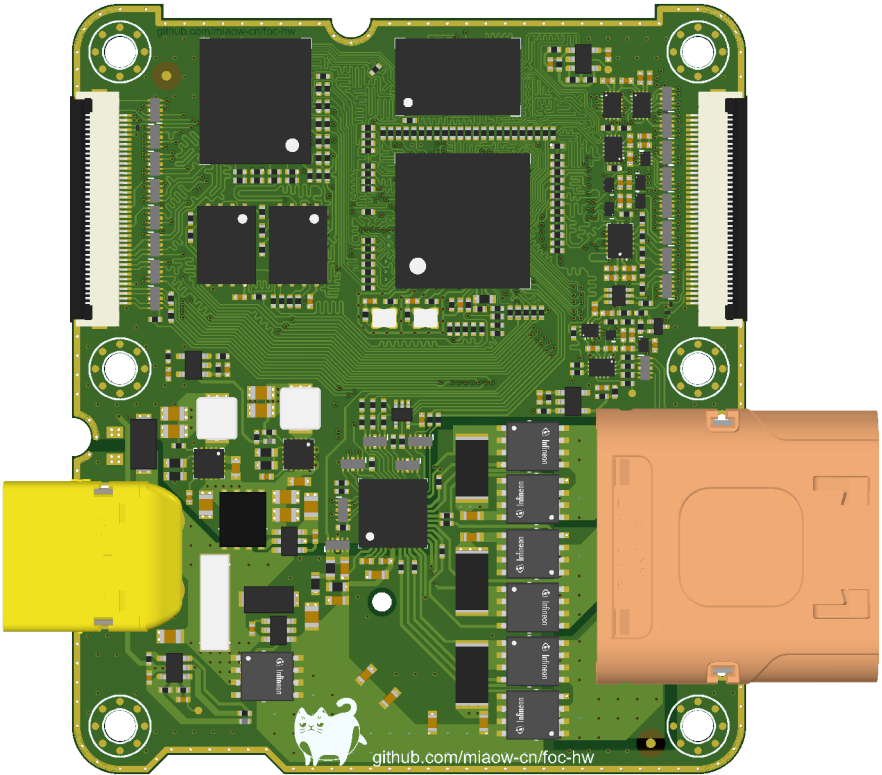


GDRVP

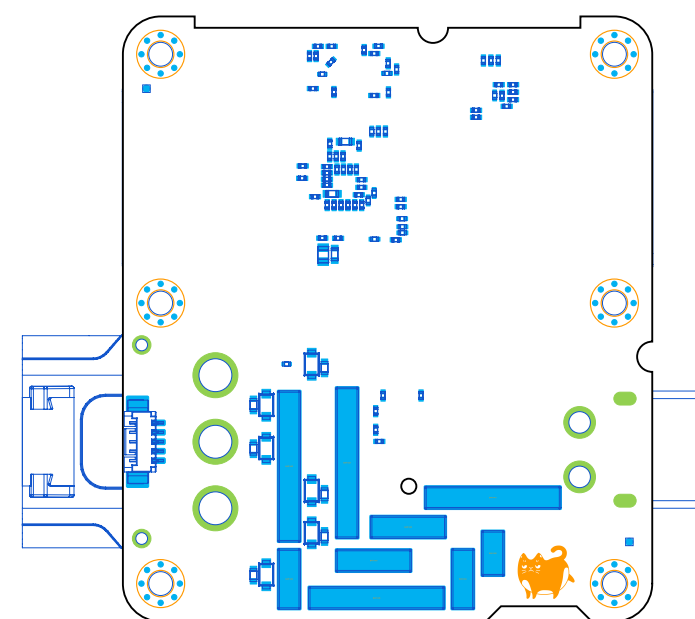
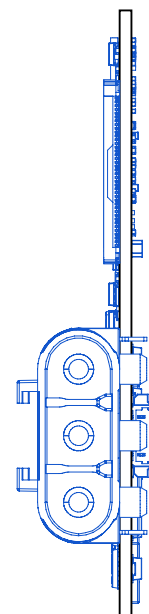
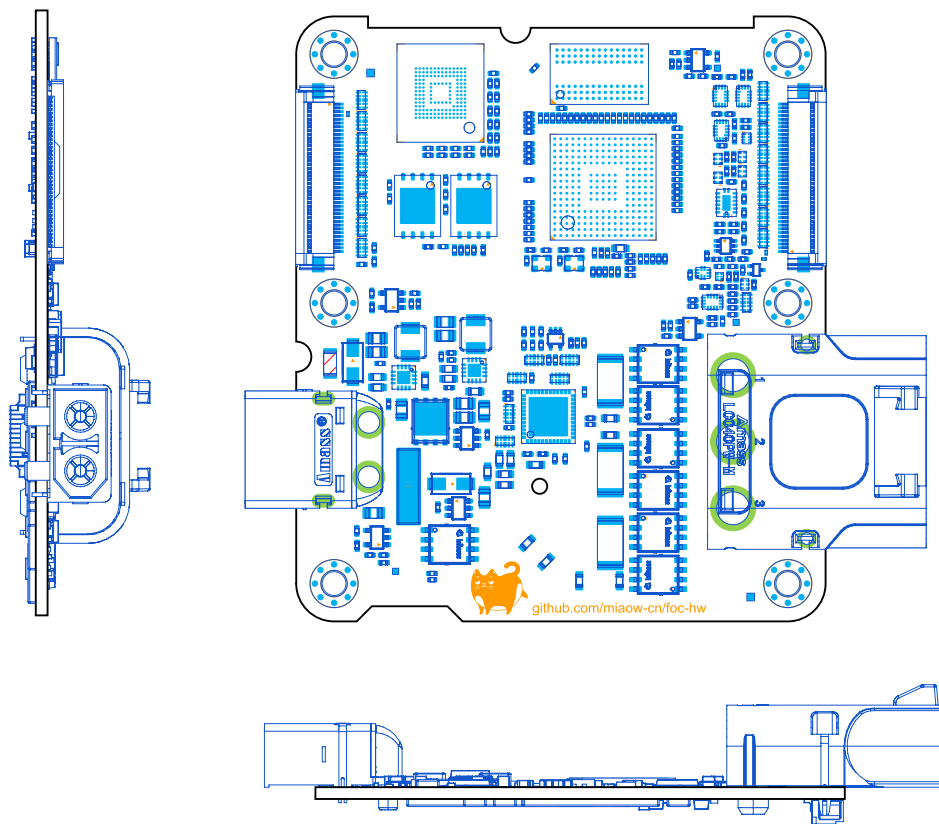


VIN

MISC Holes & marks v1.0.0 GPLv3	DRAWN10.26.2025		MIAOW FOC BOARD
	DRAWNBYmiaow		
	CHECKEDBCY		MAIN_PCB A0 
	SHEETS14		GPLv3
	SHEETS_NO14		github.com/miaow-cn/foe-hw



MIAOW realistic view	GPLv3	DRAWN	26.03.2026	MIAOW FOC BOARD		
		DRAWNBY	miaow	MAIN_PCB	A0	
		CHECKED	BCY			
		SHEETS	7	GPLv3		
		SHEETS_NO	1	github.com/miaow-cn/pcb-lib		



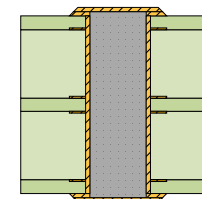
MIAOW realistic view	GPLv3	MIAOW FOC BOARD	
		DRAWN DRAWNBY CHECKED	26.03.2026 miaow BCY
		SHEETS SHEETS_NO	7 2
		MAIN_PCB A0 github.com/miaow-cn/pcb-lb	

Layer Stack Legend

Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
	Surface Material			Solder Resist	GTS
CF-004	Top Solder	0mm		Solder Mask	GTL
<i>Prepreg</i>	SIG1	0mm	<i>PP-015</i>	Signal	GTL
CF-003	GND2	0mm		Dielectric	
				Internal Plane	GP1
<i>Core</i>		1mm	<i>Core-037</i>	Dielectric	
CF-003	SIG3	0mm		Signal	G1
<i>Prepreg</i>		0mm	<i>PP-016</i>	Dielectric	
CF-003	PWR4	0mm		Internal Plane	GP2
<i>Core</i>		1mm	<i>Core-037</i>	Dielectric	
CF-003	GND5	0mm		Internal Plane	GP3
<i>Prepreg</i>		0mm	<i>PP-015</i>	Dielectric	
CF-004	SIG6	0mm		Signal	GBL
Surface Material	Bottom Solder	0mm	Solder Resist	Solder Mask	GBS
	Bottom Overlay			Legend	GBO

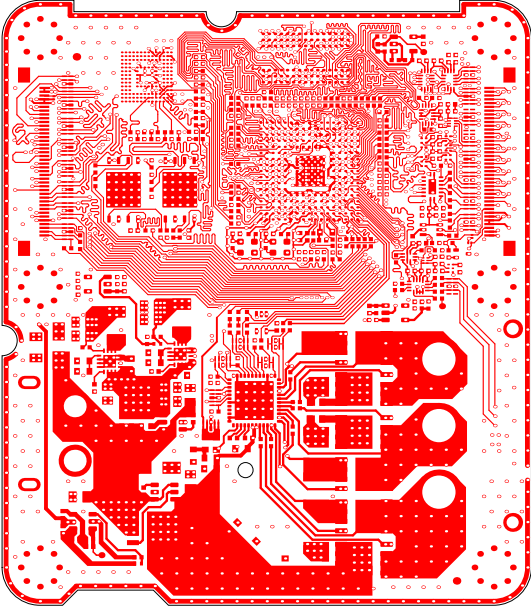
Total thickness: 2mm

Via Type Type 7 (Scale 16:1)

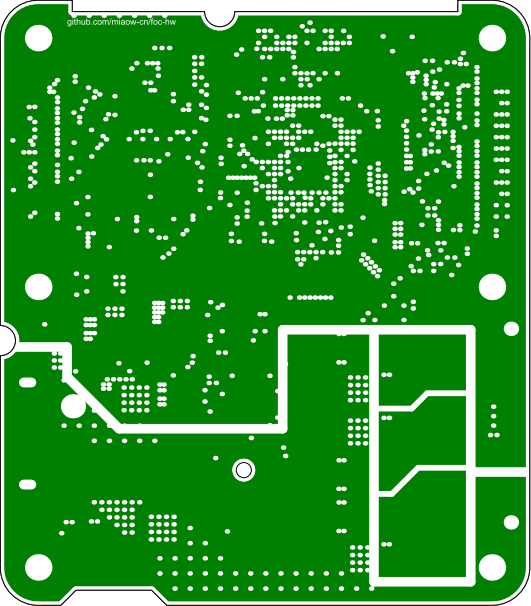


MIAOW realistic view	GPLv3	DRAWN	26.03.2026	MIAOW FOC BOARD	
		DRAWNBY	miaow		
		CHECKED	BCY	MAIN_PCB	A0
				SHEETS	7
		SHEETS_NO	3	github.com/miaow-cn/pcb-lb	

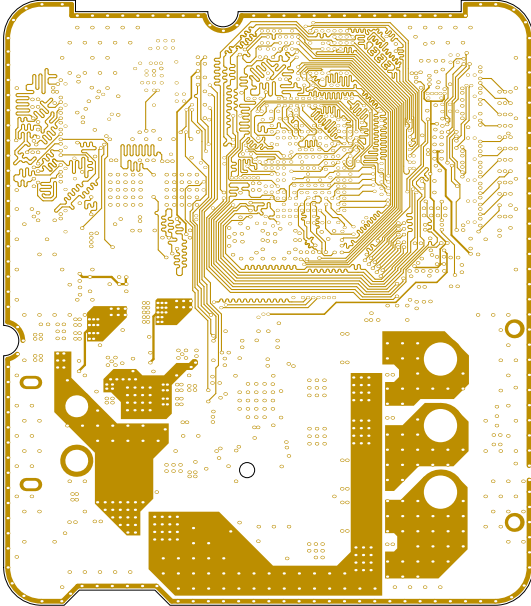
SIG1 (Scale 1:1)



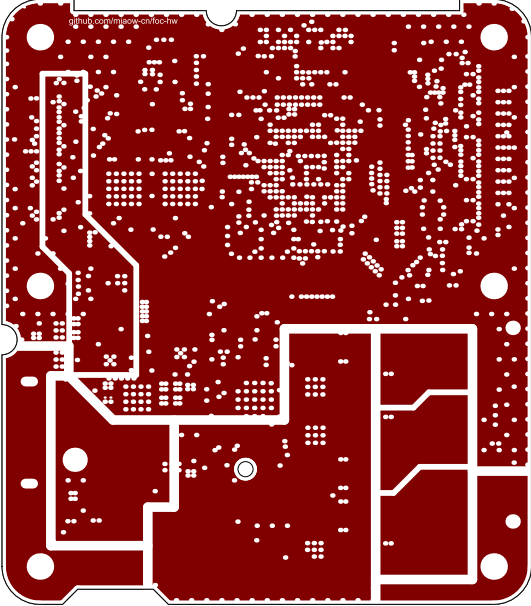
GND2 (Scale 1:1)



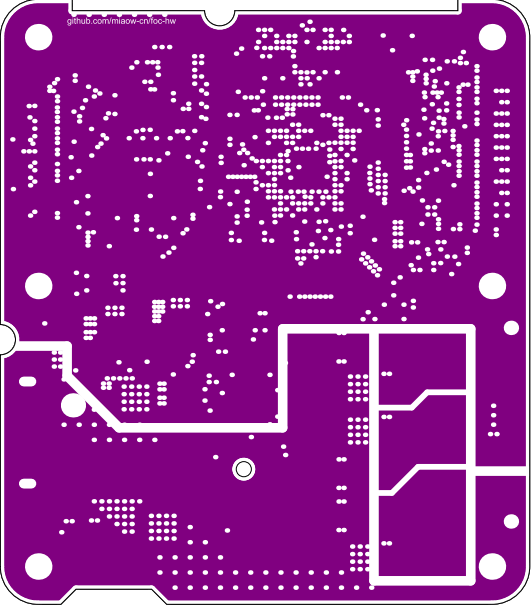
SIG3 (Scale 1:1)



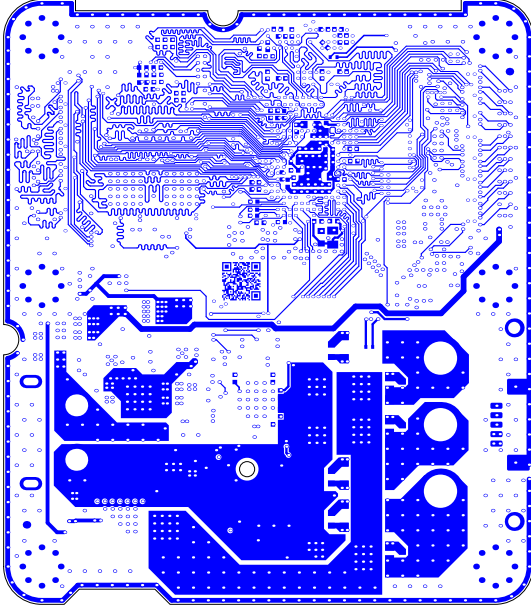
PWR4 (Scale 1:1)



GND5 (Scale 1:1)

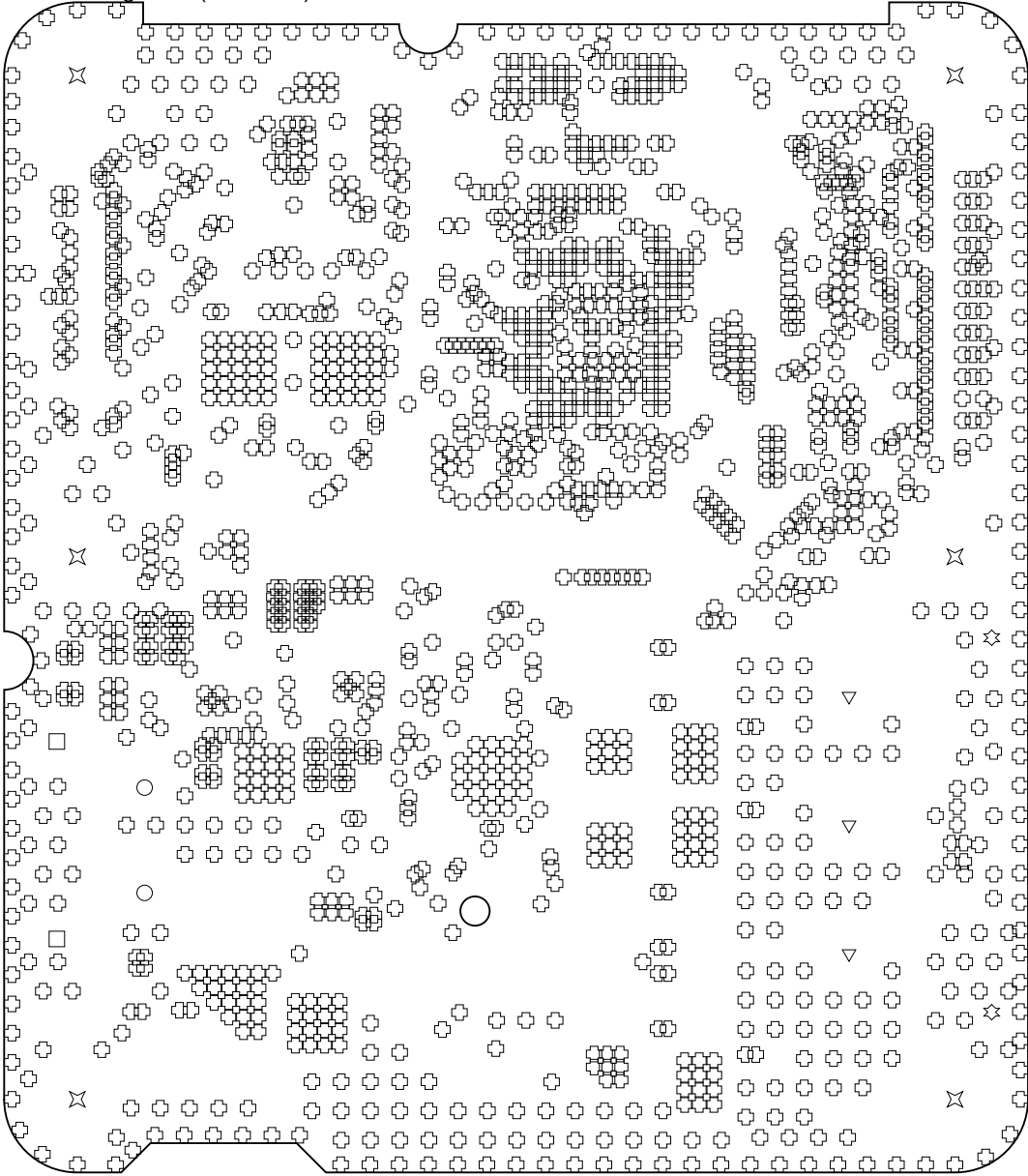


SIG6 (Scale 1:1)



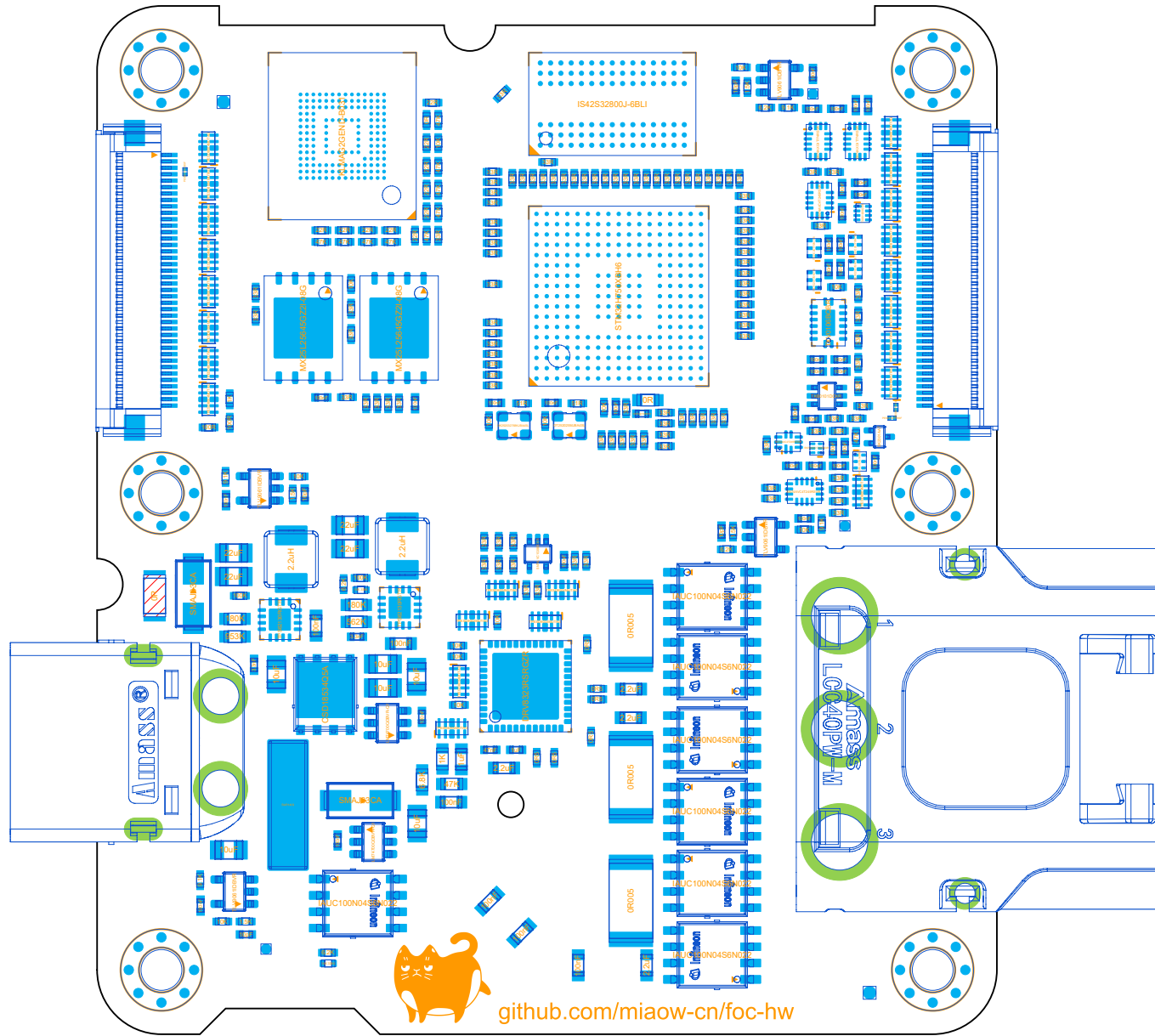
MIAOW realistic view	DRAWN26.03.2026		MIAOW FOC BOARD	
	DRAWNBY	miaow	MAIN_PCB	A0
	CHECKED	BCY		
	SHEETS	7	GPLv3	
	SHEETS_NO	4	github.com/miaow-cn/pcb-lb	

Drill Drawing View (Scale 2:1)

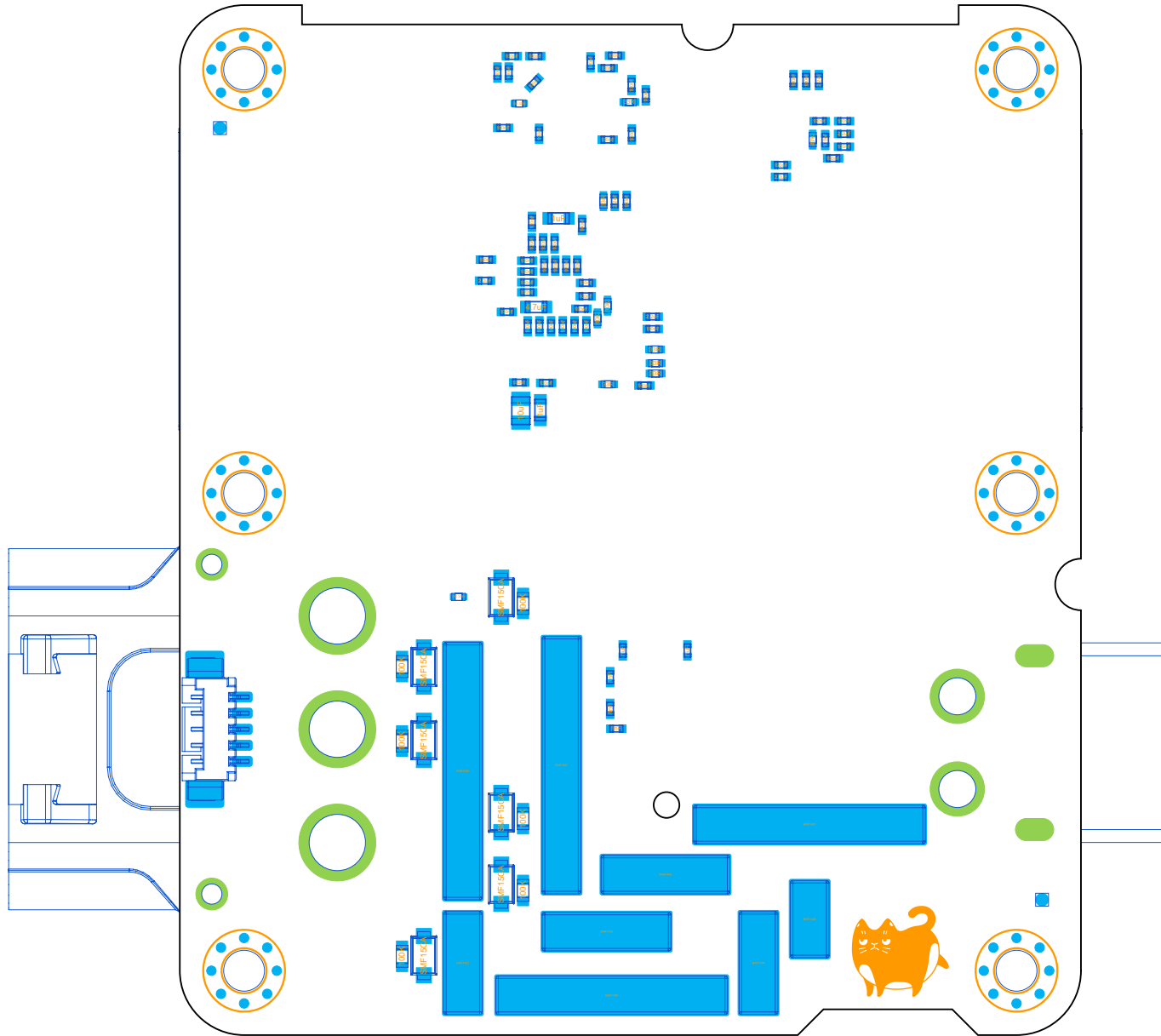


Drill Table

Symbol	Count	Hole Size	Plated	Hole Tolerance
⊕	1697	0mm	Plated	
□	2	1mm	Plated	
☆	2	2mm	Non-Plated	
○	2	3mm	Plated	
✱	6	3mm	Non-Plated	
▽	3	4mm	Non-Plated	
1712 Total				



MIAOW realistic view	GPLv3	MIAOW FOC BOARD	
		DRAWN DRAWNBY CHECKED	26.03.2026 miaow BCY
		SHEETS SHEETS_NO	MAIN_PCB A0
		7 6	GPLv3 github.com/miaow-cn/pcb-lb



MIAOW realistic view	GPLv3	DRAWN	26.03.2026	MIAOW FOC BOARD	
		DRAWNBY	miaow		
		CHECKED	BCY	MAIN_PCB	A0
		SHEETS	7	GPLv3	
		SHEETS_NO	7	github.com/miaow-cn/pcb-ib	



github.com/miaow-cn/pcb-lab